Practice problems (1)

Assumptions: All cash flows are risk free, and the interest rate is 10% (unless otherwise stated)

Question 1. You need money to go to the movies in period 2. There are two possible jobs:

- Job A pays $40 in period 1, and job B pays $55 in period 3
- **Which job should you take? How much money can you spend at the movies?**

Question 2. You want to work in periods 1, 2, and 3. You have a choice of two jobs, which pay as follows:

- Job A pays: $5,500, $4,840, and $3,993
- Job B pays: $3,300, $7,260, and $2,662
- **Which job pays better?**

Question 3. You are raising money from investors for your start-up, which you plan to sell for $58,564 at time 4. The investors will be repaid with the proceeds from the sale, at time 4.

- **What is the maximum amount of capital you can you raise from investors today?**

Question 4. Ten years ago, you invested in the stock market, and today the value of the investment has doubled.

- **Assuming a constant return, what has been your annual return?**

Question 5. You have saved $5,000 for your retirement, in 50 years. You have a choice between two savings accounts for investing, both paying 10% annual interest. The first account charges no fees, and the second one charges an annual fee of just 0.50% (leaving 9.50% in net interest)

- **How much will you have at retirement with each of the two accounts?**

Practice problems: Answers

- **Question 1:** You need money to go to the movies in period 2. There are two possible jobs:
  - **ANSWER:** Job A pays $40 in period 1 (PV = $40/1.1 = $36.4, compound forward 2 periods = $44)
  - Job B pays $55 in period 3 (PV = $55/1.1^3 = $41.3, compound forward 2 periods = $50)
  - Alternative solution, take job A and place $40 in bank for one period versus take job B borrow against future earnings for one period, leaving you with $50
  - **POINT: If you want to spend at a certain point in time, you can move money there, and the greater the PV, the more money you can have then**

- **Question 2:** You work for three periods (periods 1, 2, and 3) and have a choice between two jobs. Each job pays as follows in the three periods:
  - **ANSWER:** Job A pays: $5,500, $4,840, and $3,993 (PV = $5,000, $4,000, $3,000; sum(PV) = $12,000
  - Job B pays: $3,300, $7,260, and $2,662 (PV = $3,000, $6,000, $2,000; sum(PV) = $11,000)
  - **POINT: Mechanics of multi-period PV calculation**

- **Question 3:** You are raising money for your start-up, which you expect to be sold for $58,564 in period 4. How much money can you raise from an investor?
  - **ANSWER:** $58,654 /(1.10)^4 = $40,000
  - **POINT:** Introduce opportunity cost of capital, investors need to earn a competitive return, a company needs to deliver this return to investors

Practice problems: Answers

- **Question 4:** Ten years ago, you invested in the stock market and today the value of the investment has doubled. What is the corresponding annual return on your investment?
  - **ANSWER:** Dividing out initial investment in PV equation implies that we should solve the equation: $(1+r)^{10} = 2$. Take 10^{th} root to solve for $r = 7.18\%$ annual return.
  - **POINT:** "Rule of 72" is a useful rule of thumb. It says that the doubling time for an investment earning rate r is approximately: $time = 72 / r$ (or equivalently: $r = 72 / time$) .
  - This is also known as the "Compound Annual Growth Rate" (CAGR)

- **Question 5:** You have saved $5,000 for your retirement 50 years from now (in period 50). You have a choice between two savings accounts, the first one pays 10.00% interest and the other pays 9.50% interest. How much money will you have when you retire, if you put your $5,000 in each account?
  - **ANSWER:** Account A: $5,000 x 1.10^50 = $586,954 (117x). Account B: $5,000 x 1.095^50 = $467.386 (93x)
  - **POINT:** Power of long-term compounding, importance of small fee/return differences over long horizons, (93/117= 0.796 => 20% of potential retirement saving will go to fees)